

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2014

Paper: CSB-401 — Numerical Computing

Time: Three hours

Maximum Marks: 70

Instructions: Answer five questions including Question No. 1 which is compulsory. Use of calculator is allowed. All undefined symbols have their usual meanings.

Question 1.

- (a) Find the maximum absolute error and maximum relative error in $3x^2y/z$ when $\delta x = \delta y = \delta z = 0.001$ at $x = y = z = 1$. [3]
- (b) Define Normalized floating point number. Represent 54.75×10^7 in normalized floating point mode. [3]
- (c) Prove or disprove the relations: [4]
- (i) $\Delta^n \equiv \delta^n E^{n/2}$
- (ii) $\Delta + \nabla \equiv \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta}$
- (d) Find the missing value of the following data: [3]

x	1	2	3	4	5
y	7	—	13	21	31

- (e) Using Jacobi method, find all the eigenvalues and corresponding eigenvectors of the matrix: [3]

$$A = \begin{bmatrix} 3 & \sqrt{3} \\ \sqrt{3} & 3 \end{bmatrix}$$

- (f) Using forward differences, obtain the formula for the second derivative of $y(x)$ at $x = x_0$. [3]
- (g) Use Runge-Kutta second order method to approximate y when $x = 0.1$, given that $\frac{dy}{dx} = x + y$, $y(0) = 1$. [3]

Question 2.

- (a) Solve the equation $x \tan x + 1 = 0$ by Regula-Falsi method starting with $x_0 = 2.5$ and $x_1 = 3$ correct to three decimal places. [6]
- (b) Perform four iterations of Gauss-Seidel method to solve the equations: [6]

$$x + 2y + 10z = 61$$

$$10x + y + 2z = 44$$

$$2x + 10y + z = 51$$

by taking the initial approximation as $[0, 0, 0]$.

Question 3.

(a) Obtain the order of convergence of the secant method in finding the root of the equation $f(x) = 0$. [6]

(b) From the following table of values of x and y , obtain $\frac{d^2y}{dx^2}$ at $x = 2.2$: [6]

x	1.0	1.2	1.4	1.6	1.8	2.0	2.2
y	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

Question 4.

(a) Using the iterative method, find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 10 & 1 \\ 2 & 0 & 1 \\ 3 & 3 & 2 \end{bmatrix}$$

by assuming the initial inverse of the matrix to be

$$\begin{bmatrix} 0.4 & 2.4 & -1.4 \\ 0.14 & 0.14 & -0.14 \\ -0.85 & -3.8 & 2.8 \end{bmatrix}$$

[6]

(b) By means of Newton's divided difference formula, find a polynomial and hence find the value of $f(8)$ from the following data: [6]

x	4	5	7	10	11	13
y	48	100	294	900	1210	2028

Question 5.

(a) Prove the identity: [6]

$$y_x - \Delta^2 y_x + \Delta^3 y_x - \Delta^5 y_x + \Delta^6 y_x - \Delta^8 y_x + \dots = y_x - \Delta^2 y_{x-1} + \Delta^4 y_{x-2} - \Delta^6 y_{x-3} + \Delta^8 y_{x-4} - \dots$$

(b) Solve the following differential equation using Runge-Kutta method of order four: [6]

$$\frac{dy}{dx} = xy + y^2, \quad y(0) = 1,$$

at $x = 0.1$ and 0.2 .

Question 6.

(a) Derive Newton-Cotes quadrature formula. Hence obtain Simpson's one-third rule of integration. [6]

(b) Apply Gauss-Jordan method to solve the system of linear equations: [6]

$$2x + 3y - z = 5$$

$$4x + 4y - 3z = 3$$

$$2x - 3y + 2z = 2$$

Question 7.

- (a) Using Milne's method, find $y(0.8)$ if $y(x)$ is the solution of $\frac{dy}{dx} = 1 - y^2$, given that $y(0) = 0, y(0.2) = 0.2027, y(0.4) = 0.4228, y(0.6) = 0.6844$. [6]
- (b) Derive two-point Gauss-Legendre formula for integration. [6]